

1. $P_X(k) = C/k^\alpha$, $\forall k \in \mathbb{N}$, $\alpha > 1$

$P_Y(k) = C/(k^{\alpha+0.5})$, $\forall k \in \mathbb{Z}$, $\alpha > 1$

determine for which value α , the mean and variance of X and Y are finite, infinite, or not well defined

$$E(X) = \sum_{k \in \mathbb{N}} k P_X(k) = \sum_{k=1}^{\infty} k \frac{C}{k^\alpha} = \sum_{k=1}^{\infty} \frac{C}{k^{\alpha-1}} = \begin{cases} < \infty & \alpha-1 > 1, \alpha > 2 \text{ finite} \\ \infty & \alpha-1 \leq 1, 1 < \alpha \leq 2 \text{ infinite} \end{cases}$$

$Var(X) = E(X^2) - E(X)^2$

$$E(X^2) = \sum_{k=1}^{\infty} k^2 \frac{C}{k^\alpha} = \sum_{k=1}^{\infty} \frac{C}{k^{\alpha-2}} = \begin{cases} < \infty & \alpha-2 > 1, \alpha > 3 \text{ finite} \\ \infty & \alpha-2 \leq 1, 1 < \alpha \leq 3 \text{ infinite} \end{cases}$$

$$\therefore Var(X) = \begin{cases} \text{undefined} & 1 < \alpha \leq 2 \\ \text{infinite} & 2 < \alpha \leq 3 \\ \text{finite} & \alpha > 3 \end{cases}$$

$$E(Y) = \sum_{y \in \mathbb{Z}} y \frac{C}{(y^2+0.5)} = \sum_{y=0}^{\infty} \frac{Cy}{y^2+0.5} + \sum_{y=-1}^{-\infty} \frac{Cy}{y^2+0.5}$$

$$\sum_{y=0}^{\infty} \frac{Cy}{y^2+0.5} = \begin{cases} < \infty & \alpha-1 > 1, \alpha > 2 \text{ finite} \\ \infty & \alpha-1 \leq 1, 1 < \alpha \leq 2 \text{ infinite} \end{cases}$$

$$\sum_{y=-1}^{-\infty} \frac{Cy}{y^2+0.5} = \text{undefined}$$

$\therefore E(Y)$ is undefined

$Var(Y) = E(Y^2) - E(Y)^2$ $\therefore Var(Y)$ is undefined.

2. Gubner Chapter 2, Problem 33

$X \sim \text{Geo}_0(1/2)$, Compute $E(I_{(2,6)}(X))$

$$E(I_{(2,6)}(X)) = \sum_x I_{(2,6)}(x) P_X(x) = P_X(3) + P_X(4) + P_X(5)$$

$$P_X(x) = \frac{1}{2} \left(\frac{1}{2}\right)^x = \left(\frac{1}{2}\right)^{x+1}$$

$$\therefore E(I_{(2,6)}(X)) = \sum_{x=3}^5 \left(\frac{1}{2}\right)^{x+1} = \frac{1}{16} + \frac{1}{32} + \frac{1}{64} = 7/64$$

3. $X \sim \text{Pois}(\lambda)$, compute $E(1/(X+1))$.

$$\begin{aligned} E(1/(X+1)) &= \sum_{x=0}^{\infty} \frac{1}{x+1} P_X(x) = \sum_{x=0}^{\infty} \frac{1}{x+1} \frac{e^{-\lambda} \lambda^x}{x!} = \sum_{x=0}^{\infty} \frac{e^{-\lambda} \lambda^{x+1}}{(x+1)!} \cdot \frac{1}{\lambda} \\ &= \frac{1}{\lambda} \sum_{t=1}^{\infty} \frac{e^{-\lambda} e^{\lambda}}{t!} = \frac{e^{-\lambda}}{\lambda} \left(\sum_{t=0}^{\infty} \frac{e^{\lambda}}{t!} - 1 \right) = \frac{e^{-\lambda}}{\lambda} (e^{\lambda} - 1) = \frac{1}{\lambda} - \frac{e^{-\lambda}}{\lambda} \end{aligned}$$

4. Gubner Chapter 2, Problem 38

$$E(X) = m, \text{Var}(X) = \sigma^2$$

Find the constant c minimizes the mean-squared error $E((X-c)^2)$

$$\begin{aligned} E((X-c)^2) &= E(X^2 - 2Xc + c^2) = E(X^2) - 2cE(X) + c^2 = \text{Var}(X) + E(X)^2 - 2cE(X) + c^2 \\ &= c^2 - 2mc + m^2 + \sigma^2 = (c-m)^2 + \sigma^2 \end{aligned}$$

When $c=m$ minimizes the MSE

5. Gubner Chapter 2, Problem 45.

$Y := X_1 + X_2 + X_3 + \dots + X_M$ where X_k are i.i.d. $\text{Geo}(p)$.

Find $E(Y^2)$

$$E(Y) = E(X_1 + X_2 + \dots + X_M) = E(X_1) + E(X_2) + \dots + E(X_M) = ME(X)$$

$$X \sim \text{Geo}(p), P_X(x) = (1-p)p^{x-1}$$

$$E(X) = \sum_{x=1}^{\infty} x(1-p)p^{x-1} = \sum_{x=1}^{\infty} x(1-p)p^{x-1}$$

$$\text{from } \sum_{x=1}^{\infty} p^{x-1} = \frac{1}{1-p}$$

$$\frac{d}{dp} \sum_{x=1}^{\infty} p^x = \frac{d}{dp} \left(\frac{1}{1-p} \right)$$

$$= \sum_{x=1}^{\infty} \frac{d}{dp} p^x = \sum_{x=1}^{\infty} x p^{x-1} = \frac{1}{(1-p)^2}$$

$$\therefore E(X) = \frac{1}{1-p}$$

$$\Rightarrow E(Y) = \frac{M}{1-p}$$

$$\begin{aligned}
\text{Var}(Y) &= \text{Var}(X_1 + X_2 + \dots + X_M) \quad \because X_i \stackrel{\text{iid}}{\sim} \text{Geo}(p) \rightarrow \text{Var}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2 \\
&= \text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_M) = M \text{Var}(X) \\
&= M \left(\frac{1}{(1-p)^2} - \frac{1}{(1-p)} \right)
\end{aligned}$$

$$\begin{aligned}
\mathbb{E}[X(X+1)] &= \sum_{x=0}^{\infty} x(x+1)(1-p)p^{x-1} \\
&= \sum_{x=1}^{\infty} x(x+1)(1-p)p^{x-1} \\
\sum_{x=1}^{\infty} p^{x+1} &= \frac{p^2}{1-p} \\
\Rightarrow \frac{d^2}{dp^2} \sum_{x=1}^{\infty} p^{x+1} &= \sum_{x=1}^{\infty} \frac{d^2}{dp^2} p^{x+1} = \frac{2}{(1-p)^3} \\
\Rightarrow \sum_{x=1}^{\infty} x(x+1)p^{x-1} &= \frac{2}{(1-p)^3} \\
\therefore \mathbb{E}(X^2) + \mathbb{E}(X) &= \frac{2}{(1-p)^2} \\
\text{Var}(X) &= \frac{2}{(1-p)^2} - \frac{1}{1-p} - \frac{1}{(1-p)^2}
\end{aligned}$$

6. Gubner Chapter 2, Problem 50.

X : discrete random variable taking $x_1, x_2, \dots, x_n$

$$p_i := P(X=x_i)$$

$$g(x) := -\log P(X=x)$$

Observe $g(x_i) = -\log p_i$ entropy X is defined.

$$H(X) := \mathbb{E}(g(X)) = \sum_{i=1}^n g(x_i) P(X=x_i) = \sum_{i=1}^n p_i \log \frac{1}{p_i}$$

1) if all outcomes are equally likely $p_i = 1/n$

$$H(X) = \sum_{i=1}^n \frac{1}{n} \log n = \log n$$

2) if X is constant random variables. $p_j = 1$ for some j and $p_i = 0$ for $i \neq j$

$$H(X) = \sum_{i=1}^n p_i \log \frac{1}{p_i} = p_j \log \frac{1}{p_j} + 0 = 1 \log(1) = 0$$

7. Gubner Chapter 3. Problem 38

$$P_{XY}(i, j) = \begin{cases} \frac{3j-1}{j!} e^{-3} & i=1, j \geq 0 \\ 4 \frac{6j-1}{j!} e^{-6} & i=2, j \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Compute $E(Y|X=i)$, $E(Y)$ and $E(X|Y=j)$

$$P_X(i=1) = \sum_{j=0}^{\infty} P_{XY}(i=1, j) = \frac{1}{3} \quad P_X(i=2) = \sum_{j=0}^{\infty} P_{XY}(i=2, j) = \frac{2}{3} \quad P_Y(j) = \sum_{i=1}^2 P_{XY}(i, j) \\ = \frac{1}{3} \left(\frac{e^{-3} 3j}{j!} \right) + \frac{2}{3} \left(\frac{e^{-6} 6j}{j!} \right)$$

$$E(Y|X=i) = \sum_j y \cdot P(Y=y|X=i)$$

$$E(Y|X=1) = \sum_y y P(Y=y|X=1)$$

8. Gubner Chapter 3, Problem 41

9. Gubner Chapter 3. Problem 45

10. Dobrow Problem 4.59.

Variance of matching problem

set $I_k = \begin{cases} 1 & \text{kth student gets their cap back} \\ 0 & \text{otherwise} \end{cases}$

$$E(X) = E\left(\sum_{k=1}^n I_k\right) = \sum_{k=1}^n P(I_k=1) = \sum_{k=1}^n \frac{1}{n} = 1$$

$$\text{Var}(X) = \text{Var}\left(\sum_{k=1}^n I_k\right) = \sum_{k=1}^n \text{Var}(I_k) + \sum_{i \neq j} \text{Cov}(I_i, I_j)$$

$$E(I_i I_j) - E(I_i)E(I_j)$$

$$= \frac{1}{n} \cdot \frac{1}{n-1} - \frac{1}{n} \cdot \frac{1}{n} = \frac{1}{n} \left(\frac{1}{n-1} - \frac{1}{n} \right)$$

$$E(I_k^2) - E(I_k)^2 = \frac{1}{n} - \frac{1}{n^2} = \frac{1}{n} \left(1 - \frac{1}{n} \right)$$

$$\therefore \text{Var}(X) = \sum_{k=1}^n \frac{1}{n} \left(1 - \frac{1}{n} \right) + \sum_{i \neq j} \frac{1}{n} \left(\frac{1}{n-1} - \frac{1}{n} \right)$$

$$= \left(1 - \frac{1}{n} \right) + n(n-1) \frac{1}{n} \left(\frac{1}{n-1} - \frac{1}{n} \right) = 1 - \frac{1}{n} + 1 - \frac{n-1}{n} = 1$$